

Requirement Analysis Report

Document Information

Field	Value
Report ID	SCRUM-101-REQ-ANALYSIS
Source Document	user_stories/SCRUM-101-ecommerce-checkout.md
Source Document Title	SCRUM-101 - E-commerce Checkout Process
Document Type	User Story
Application Under Test	SauceDemo (https://www.saucedemo.com)
Test Credentials Provided	standard_user / secret_sauce
Analyzed By	Requirement Analysis Agent
Analysis Date	2026-09-17
Report Version	1.2 (re-run — regenerated from unchanged source input, no feedback provided)

Business Objective

Enable a logged-in customer to complete an online purchase through a checkout process that lets them review the cart, enter shipping information, review order/payment details, and confirm the order — with the process being intuitive, secure, and providing clear feedback at every step.

Requirement Summary

The story defines a four-stage checkout flow (Cart Review → Checkout Information Entry → Order Overview → Order Completion), gated behind login and a non-empty cart, with mandatory-field validation and error handling on the information-entry step. Five acceptance criteria (AC1–AC5) and five business rules (BR1–BR5) are documented. Several implementation details relevant to QA scope — field validation formats, security controls, tax calculation, and cancel behavior on the information step — are not specified in the source document and are logged as Open Questions rather than assumed.

Functional Requirements

ID	Requirement	Source
FR1	The system shall display all cart items with name, description, price, and quantity on the cart page.	AC1
FR2	The system shall display the total price calculation on the cart page.	AC1
FR3	The system shall provide options to continue shopping or proceed to checkout from the cart page.	AC1
FR4	The system shall redirect the user to the checkout information page when the "Checkout" button is clicked.	AC2
FR5	The system shall display form fields for First Name, Last Name, and Zip/Postal Code on the checkout information page.	AC2

ID	Requirement	Source
FR6	The system shall treat all checkout information fields as mandatory.	AC2, BR1
FR7	The system shall display an error message indicating which field is required when a mandatory field is left empty and "Continue" is clicked.	AC2
FR8	The system shall redirect to the checkout overview page when "Continue" is clicked with valid checkout information.	AC3
FR9	The system shall display a summary of all items in the order on the overview page.	AC3
FR10	The system shall display payment and shipping information on the overview page.	AC3
FR11	The system shall display subtotal, tax, and total amount on the overview page.	AC3
FR12	The system shall provide options to Cancel or Finish the order from the overview page.	AC3
FR13	The system shall redirect to the order confirmation page when "Finish" is clicked.	AC4
FR14	The system shall display a success message confirming the order on the confirmation page.	AC4
FR15	The system shall display a "Back Home" button on the confirmation page to return to the products page.	AC4
FR16	The system shall display validation error messages when invalid data (special characters, incomplete information) is entered on the checkout information page.	AC5
FR17	The system shall prevent the user from proceeding until all checkout information fields are valid.	AC5
FR18	The system shall restrict access to checkout to logged-in users only.	BR2
FR19	The system shall prevent proceeding to checkout when the cart is empty.	BR3
FR20	The system shall clear the cart upon order confirmation.	BR4
FR21	The system shall allow the user to cancel checkout at any step and return to the cart.	BR5

Non-Functional Requirements

ID	Requirement	Category	Source
NFR1	The checkout flow shall be tested/supported across Chrome, Firefox, and Safari browsers.	Compatibility	Technical Notes
NFR2	The checkout flow shall be mobile-responsive.	Usability / Responsiveness	Technical Notes
NFR3	The checkout process shall be "secure" (no concrete controls specified).	Security	Story Description
NFR4	The checkout process shall be "intuitive" and provide "clear feedback" at each step.	Usability	Story Description
NFR5	All checkout form validation messages shall be validated as part of testing.	Reliability / Testability	Technical Notes

> Note: NFR3 and NFR4 are stated qualitatively in the source with no measurable acceptance threshold. See Open Questions.

Actors

Actor	Description	Source
Customer / Logged-in User	Primary actor; initiates and completes the checkout flow. Represented in testing by the <code>standard_user</code> test credential.	Story Title, AC1–AC5, BR2
SauceDemo Application (System)	The system under test that renders cart, checkout, and confirmation pages and enforces validation.	Implied throughout

No additional roles (e.g., admin, guest checkout, external payment provider) are documented.

Acceptance Criteria

Preserved verbatim from the source document.

AC1: Cart Review

- GIVEN I am a logged-in user with items in my cart
- WHEN I navigate to the cart page
- THEN I should see all added items with their details (name, description, price, quantity)
- AND I should see the total price calculation
- AND I should have options to continue shopping or proceed to checkout

AC2: Checkout Information Entry

- GIVEN I am on the cart page with items
- WHEN I click the "Checkout" button
- THEN I should be redirected to the checkout information page
- AND I should see form fields for First Name, Last Name, and Zip/Postal Code
- AND all fields should be mandatory
- WHEN I leave any field empty and click Continue
- THEN I should see an error message indicating which field is required

AC3: Order Overview

- GIVEN I have entered valid checkout information
- WHEN I click the "Continue" button
- THEN I should be redirected to the checkout overview page
- AND I should see a summary of all items in my order
- AND I should see payment and shipping information
- AND I should see the subtotal, tax, and total amount
- AND I should have options to Cancel or Finish the order

AC4: Order Completion

- GIVEN I am on the checkout overview page
- WHEN I click the "Finish" button
- THEN I should be redirected to the order confirmation page
- AND I should see a success message confirming my order
- AND I should see a "Back Home" button to return to the products page

AC5: Error Handling

- GIVEN I am on the checkout information page
- WHEN I enter invalid data (e.g., special characters, incomplete information)
- THEN I should see appropriate validation error messages
- AND I should not be able to proceed until all fields are valid

Business Rules

ID	Rule
BR1	All checkout form fields are mandatory
BR2	Users must be logged in to access checkout
BR3	Cart cannot be empty when proceeding to checkout
BR4	Order confirmation should clear the cart
BR5	Users can cancel checkout at any step and return to cart

Dependencies

ID	Dependency	Notes
DEP1	User authentication / login feature	BR2 requires login before checkout access; the login flow itself is not specified in this document.
DEP2	Cart / product catalog feature	AC1 assumes items already exist in the cart; the add-to-cart flow is out of scope of this story.
DEP3	SauceDemo test environment and credentials	Testing depends on availability of https://www.saucedemo.com and the <code>standard_user</code> / <code>secret_sauce</code> account.
DEP4	Playwright automation framework and multi-browser test infrastructure	Required per Technical Notes to execute automated checks across Chrome, Firefox, and Safari.

Assumptions

ID	Assumption	Rationale
A1	"Logged-in user" refers to authenticating with the provided <code>standard_user</code> / <code>secret_sauce</code> credentials.	These are the only credentials the document supplies.
A2	The checkout flow implemented in the SauceDemo application maps to: Cart page → Checkout: Your Information → Checkout: Overview → Checkout: Complete, matching AC1–AC4.	Standard SauceDemo behavior; not explicitly re-described step-by-step beyond the ACs.
A3	No real payment is processed; "payment information" shown on the overview page (AC3) is static/demo data rather than user-entered payment details.	The document defines no payment-entry step or payment form fields anywhere in the checkout flow.
A4	Only one currency and no discount/coupon/promo-code logic is in scope.	Not mentioned anywhere in the story.

Risks

ID	Risk	Impact
R1	No Cancel control is documented for the checkout information page (AC2), yet BR5 states cancellation is possible "at any step."	Ambiguous scope could cause a missed requirement or an untestable acceptance criterion.

ID	Risk	Impact
R2	No validation rules (allowed characters, length, format) are defined for First Name, Last Name, or Zip/Postal Code.	Negative/format test cases cannot be fully specified without guessing pass/fail boundaries.
R3	"Secure" checkout (Story Description) has no concrete, testable security requirement attached.	Security testing scope is undefined; real security gaps could go unverified.
R4	Tax calculation rule/rate is not documented, despite AC3 requiring the tax amount to be displayed.	Correctness of the displayed tax cannot be independently verified.
R5	Expected behavior for browser back/forward navigation mid-checkout is not defined, even though Technical Notes call for testing it.	Testers/automation would have to guess the "correct" behavior, risking false failures or missed defects.
R6	SauceDemo is a public demo site with intentionally seeded bugs on certain accounts (e.g., alternate usernames not referenced here).	If test data/accounts change, results may reflect seeded demo bugs rather than genuine defects — out of scope for this story since only <code>standard_user</code> is specified.
R7	No performance, load, or response-time criteria are defined for the checkout flow.	Non-functional test coverage for performance cannot be planned against a measurable target.

Edge Cases

- Attempting to proceed to checkout with an empty cart (BR3).
- Leaving exactly one of First Name, Last Name, or Zip/Postal Code empty (each field individually), vs. leaving all empty.
- Entering special characters, numeric-only, or oversized input into name fields.
- Entering invalid/malformed Zip/Postal Code formats.
- Canceling checkout from the overview page and verifying cart contents are preserved.
- Canceling checkout from the information page (if such a control exists — see Open Question OQ2) and verifying cart contents are preserved.
- Verifying the cart is empty after order completion, then attempting to check out again.
- Multiple line items with varying quantities, verifying subtotal/tax/total arithmetic on the overview page.
- Using the browser back button after reaching the confirmation page, then attempting to re-finish the order.
- Mobile-viewport execution of the full checkout flow (Technical Notes).
- Session handling if login expires mid-checkout (not addressed in the source document — see Open Question OQ7).

Open Questions

ID	Question	Related Item
OQ1	What are the exact validation rules (allowed characters, min/max length, format) for First Name, Last Name, and Zip/Postal Code?	AC5, FR16, FR17
OQ2	Does the checkout information page (AC2) expose a Cancel control, given BR5 says cancellation is possible "at any step" but AC2 lists no Cancel option?	AC2, BR5
OQ3	What concrete security controls define a "secure" checkout process (e.g., transport encryption, no storage of payment data, session protections)?	Story Description, NFR3
OQ4	What tax calculation rule or rate should the overview page apply, and how should it be verified?	AC3, FR11
OQ5	Is payment method selection/entry part of this flow, or is the "payment information" on the overview page pre-populated/static demo data?	AC3, FR10
OQ6	What is the expected behavior of the browser back/forward buttons at each checkout step?	Technical Notes

ID	Question	Related Item
OQ7	What happens if the user's session/login expires while mid-checkout?	BR2
OQ8	What specific devices/viewports/breakpoints define "mobile responsiveness" for the checkout flow?	Technical Notes, NFR2
OQ9	Does canceling checkout guarantee the cart is preserved unchanged, or could cancellation alter cart contents?	BR5, FR21

Requirement Traceability

Item	Type	Source Section
FR1–FR3	Functional Requirement	AC1: Cart Review
FR4–FR7	Functional Requirement	AC2: Checkout Information Entry
FR8–FR12	Functional Requirement	AC3: Order Overview
FR13–FR15	Functional Requirement	AC4: Order Completion
FR16–FR17	Functional Requirement	AC5: Error Handling
FR18	Functional Requirement	Business Rule 2
FR19	Functional Requirement	Business Rule 3
FR20	Functional Requirement	Business Rule 4
FR21	Functional Requirement	Business Rule 5
NFR1, NFR5	Non-Functional Requirement	Technical Notes
NFR2	Non-Functional Requirement	Technical Notes
NFR3, NFR4	Non-Functional Requirement	Story Description
Customer / Logged-in User	Actor	Story Title, AC1–AC5, BR2
AC1–AC5	Acceptance Criteria	Acceptance Criteria section
BR1–BR5	Business Rule	Business Rules section
DEP1	Dependency	BR2 (inferred)
DEP2	Dependency	AC1 (inferred)
DEP3	Dependency	Application URL / Test Credentials
DEP4	Dependency	Technical Notes
A1–A4	Assumption	Inferred from Application URL, Test Credentials, AC3, Technical Notes
R1–R7	Risk	Inferred from gaps across AC2, AC3, Story Description, Technical Notes
OQ1–OQ9	Open Question	Inferred from gaps across AC2, AC3, BR5, Story Description, Technical Notes

QA Coverage Recommendations

1. Positive-path coverage for the full checkout flow: Cart Review → Checkout Information → Order Overview → Order Completion (AC1–AC4), as a smoke/regression test.
2. Mandatory-field validation coverage for AC2: each of First Name, Last Name, Zip/Postal Code left empty individually, in pairs, and all together.
3. Negative/format validation coverage for AC5: special characters, whitespace-only input, and boundary-length input — pending clarification in OQ1 before exact pass/fail rules can be finalized.
4. Cart-state verification: total price accuracy on the cart page (AC1) and subtotal/tax/total accuracy on the overview page (AC3), across single-item and multi-item/multi-quantity carts.
5. Business-rule negative tests: attempt checkout with an empty cart (BR3) and attempt to access checkout while logged out (BR2).
6. Cancel-flow coverage: cancel from the overview page (AC3) and, pending OQ2, cancel from the information page — verify cart contents are preserved in both cases (BR5, OQ9).
7. Post-completion state check: verify the cart is cleared after order confirmation (BR4) and that "Back Home" (AC4) returns to the products page correctly.
8. Cross-browser execution of the above on Chrome, Firefox, and Safari (NFR1).
9. Mobile-responsiveness pass of the full checkout flow (NFR2), scope pending OQ8.
10. Exploratory testing around browser back/forward navigation during checkout (Technical Notes, OQ6) and around the undefined "secure"/"intuitive" qualities (NFR3, NFR4, OQ3) — flag findings rather than asserting pass/fail against an undocumented standard.
11. Do not write automated assertions against tax calculation values, Zip/Postal Code format limits, or Cancel destination from the information page until OQ1, OQ2, and OQ4 are resolved with the product owner.